ASSIGNMENT 2025/2026

BSc (Hons) Actuarial Studies

SH309

Module Name: Statistical Methods

Module Code: STAT 1011Y(1)

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ASSIGNMENT 2025/2026 STATISTICAL METHODS

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Assignment Tokens

- **TOKEN-A (Portfolio):** 44371
- **TOKEN-B(Counts)**: 2406

Executive Summary

PCA led to the identification of three major risk drivers within the policy-portfolio. PC1 focuses on the Behavioural Risk, representing driver's historical reliability while PC2 explores the Exposure Risk, analysing the physical intensity of vehicle usage. PC3 (Life Stage/Asset Quality) pinpoints a strong negative correlation: high scores indicate older drivers with older vehicles and vice versa. As supported by the findings, the following actions are recommended:

- **Targeted Pricing :** Implement a 15% premium loading for high mileage vehicles. This links insurance pricing to higher depreciation and frequency of accidents observed in this portfolio, potentially increasing revenue by MUR 1.8M.
- **Risk Mitigation :** Offer telematics-based insurance for younger drivers with new vehicles, cutting down on losses and save about MUR 750,000.
- **Loyalty Incentives :** Offer a 5% discount to drivers with high NCD to maintain customer loyalty, protecting MUR 500,000 in stable income.

The results of the Chi-Square test indicates that there is significant evidence of a relationship between claim_type and transmission type at 5% level of significance driven mostly by Windscreen claims. The Poisson distribution was not suitable to show the dispersion of daily claim counts, implying that the data required another model due to overdispersion.

Problem 1: Principal Component Analysis

Introduction

The objective of this project is to uncover and interpret key components that influence policyholder risk profiles. Using Principal Component Analysis, the complexity of the data will be reduced through dimension reduction and this will enable the interpretation of hidden patterns and risk structures within the data. In turn, this supports actuarial decision making in pricing, underwriting and risk classification. The aim is to uncover emerging risk groups within the portfolio, providing actionable insights into the characteristics of policyholders.

Exploratory Data Analysis

First, we must conduct an exploratory data analysis to check whether all the observations are numerical, verify for missing values if whether we have to remove or impute data, have a summary of our dataset with various descriptive statistics to enable the detection of any questionable or extreme records that might influence our analysis and eventually determine whether there is any relationship between the variables.

Data Type Determination

We can observe that the dataset contains both numerical and categorical data by analysing the summary of data. Categorical variables namely region, fuel type, transmission and claim type should be excluded from the PCA analysis as it works only with numerical variables.

(Refer to Appendix A)

Identification of missing values

Missing values are a frequent occurrence in datasets and can often hinder the process of data analysis. Therefore, it is important to identify them and minimise the risk of bias in the analysis. Claim_severity has 1092 missing values, but this should not be flagged as it represents the number of policyholders who did not make a claim during this period. Furthermore, there are missing values in two of the main risk variables, namely, annual_mileage with 33 missing records and engine_power_kw with 15. For PCA to be conducted, imputation should be carried out, so that we have a complete dataset, that is, they must be replaced with a reliable and appropriate value, which in our case is the median values of each variable.

Upon imputation, the new values for the 2 variables were as follows:

Variable	Missing Count	Imputation Value	Imputation Type
annual_mileage	33	16143.0	Median
engine_power	15	89.9	Median

33 missing values of annual_mileage were replaced by 16 143 and the 15 missing values of engine power were replaced by 89.9kW.

After imputation, it was concluded that no missing values remained in the dataset and the imputed data needs to be standardized.

Figure 1.1 below provides a visual representation of the dataset through boxplots. This visual analysis confirms that there is substantial disparity between the measurement scales of the numerical variables. For example, Claim Severity is measured in magnitudes of thousands (MUR) while demographic metrics like Vehicle Age operate in single digits. As a result, standardisation is required to neutralise the effect of scale differences among variables so that PCA is not disproportionately biased towards high-magnitude variables. By transforming the data into Z-scores, that is, mean=0 and standard deviation=1

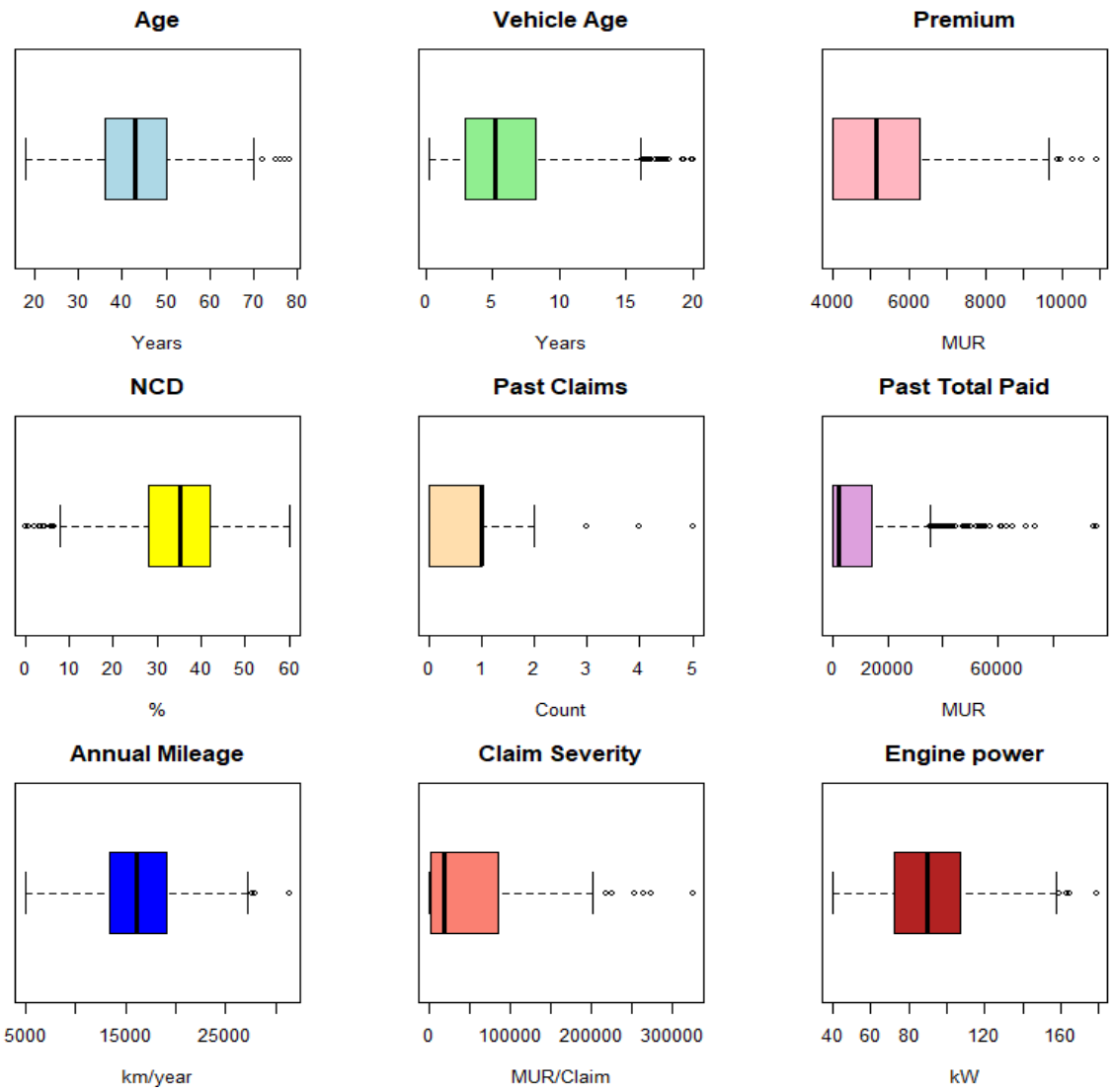

Figure 1.1: Boxplots showing outliers

Summary of Original Data

Variable	Min.	1st Qu	Median	Mean	3rd Qu	Max.
Age	18.00	36.00	43.00	42.64	50.00	78.00
Vehicle Age	0.200	2.900	5.200	6.048	8.200	20.000
Premium	4000	4000	5128	5366	6277	10903
No Claims Discount	0.00	28.18	35.30	34.64	41.92	60.00
Past Claims Count	0.0000	0.0000	1.0000	0.7233	1.0000	5.0000
Past Total Paid	0	0	2462	8864	14138	95666
Annual Mileage	5000	13427	16143	16266	19049	31330
Engine Power	40.00	72.40	89.90	90.24	106.90	178.80

Summary of Normalised (and Imputed) Data

Variable	Min.	1st Qu	Median	Mean	3rd Qu	Max.
Age	-2.3858	-0.61821	0.03486	0	0.71264	3.42378
Vehicle Age	-1.4112	-0.7597	-0.2047	0	0.5192	3.3665
Premium	-1.0188	-1.0188	-0.1777	0	0.6797	4.1308
No Claims Discount	-3.31441	-0.61821	0.06361	0	0.69759	2.42727
Past Claims Count	-0.8395	-0.8395	0.3211	0	0.3211	4.9635
Past Total Paid	-0.6948	-0.6948	-0.5018	0	0.4134	6.8046
Annual Mileage	-2.79342	-0.68407	-0.02972	0	0.66523	3.73697
Engine Power	-2.06626	-0.71909	-0.01363	0	0.68155	3.64327

The tables above provide meaningful insights into the importance of standardisation as a pre-processing step. PCA generates a new projection of the dataset, where the new axes are determined by the standard deviation of the variables. Therefore, by standardising, all variables are given equal standard deviation of 1, ensuring that each variable has the same weight and this allows PCA to compute meaningful axes.

Check for correlation

After standardization, we must identify any correlation between the numerical variables. A correlation heatmap, as shown in Fig 1.2, is useful to have a clear visual understanding of any relationship that may exist.

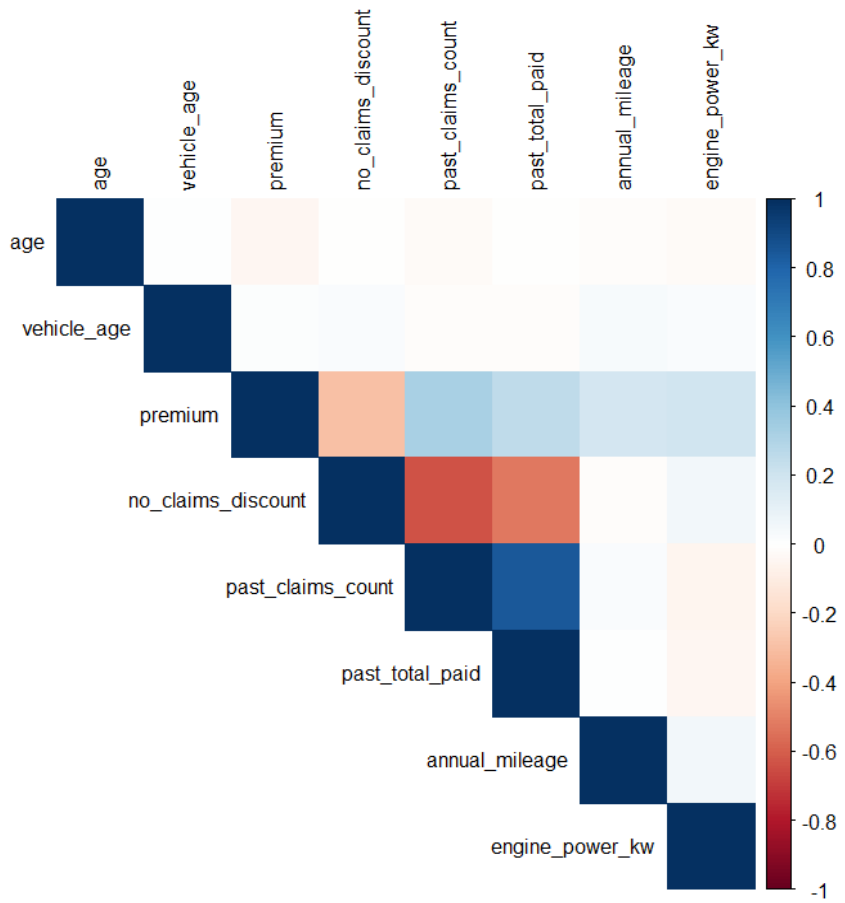

Fig1.2: Correlation Heatmap

Based on Figure 1.2, the darker the blue, the stronger the positive relationship while the darker the red, the stronger the negative relationship.

past_claims_count vs. past_total_paid (0.843): This is the strongest positive relationship in the plot. This means that the more claims a person has filed, the higher the total amount the insurance company has paid to them.

no_claims_discount vs. past_claims_count (-0.632): This is the strongest negative correlation, which is exactly what we expect in insurance. As the number of claims increases, the discount (NCD) decreases since the risk associated with such policyholders also increases.

Correlation matrix

The exact values of the other correlation values can be found summarized in the table below:

	Age	Vehicle Age	Premium	No Claims Discount	Past Claims Count	Past Total Paid	Annual Mileage	Engine Power kW
Age	1	0.005	-0.049	-0.003	-0.021	-0.01	-0.014	-0.022
Vehicle Age	0.005	1	0.014	0.024	-0.016	-0.019	0.031	0.02
Premium	-0.049	0.014	1	-0.296	0.323	0.258	0.183	0.19
No Claims Discount	-0.003	0.024	-0.296	1	-0.632	-0.525	-0.019	0.058
Past Claims Count	-0.021	-0.016	0.323	-0.632	1	0.843	0.023	-0.056
Past Total Paid	-0.01	-0.019	0.258	-0.525	0.843	1	0.002	-0.047
Annual Mileage	-0.014	0.031	0.183	-0.019	0.023	0.002	1	0.051
Engine power kW	-0.022	0.02	0.19	0.058	-0.056	-0.047	0.051	1

The correlation matrix above supports the analysis shown in Fig 1.2. We observe that Past Total Paid and Past Claims Count have the strongest positive correlation of 0.843 while Past Claims Count and No Claims Discount have the strongest negative correlation of -0.632.

Application of PCA

With reference to the table above and the heatmap in Fig 1.2, we can point out that there are correlated data in the dataset. Consequently, through the application of Principal Component Analysis, these correlated variables will be converted to smaller set of uncorrelated principal components through dimension reduction. This will allow for a more effective interpretation of the risk profiles and underlying patterns required for pricing and underwriting decisions.

After applying PCA, the data has been reduced to 8 uncorrelated principal components. This method enables us to capture the maximum variance in the portfolio and observe broad risk structures rather than inspecting individual metrics like vehicle power (kW) independently.

Summary of PCA result

	Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6	Comp.7	Comp.8
Standard Deviation	1.590	1.116	1.004	0.990	0.972	0.810	0.708	0.380
Proportion of Variance	0.316	0.156	0.126	0.122	0.118	0.0819	0.0627	0.0180
Cumulative Proportion	0.316	0.471	0.597	0.719	0.837	0.919	0.982	1.000

Next, we move to **component selection**. The selection of the optimal number of components is a critical step as there are risks of underfitting or overfitting the model. If a small number of components are chosen, we may miss out on relevant data in the analysis. Conversely, retaining too many components introduces 'noise' which makes it trickier to interpret risk profiles.

In PCA, eigenvalues represent the amount of variance (information) captured by each principal component. Since we started with 8 variables and standardised them, the total variance in our dataset is exactly 8.0. The values computed for the eigenvalues are as follows:

PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8
2.525	1.245	1.007	0.978	0.944	0.655	0.501	0.144

1. **The Kaiser Criterion (The "Rule of 1")**

The Kaiser's Rule (Kaiser-Guttman criterion) is a numerical threshold which is used to decide which principal components are important enough to keep. The most common rule for selecting components is to keep any component with an eigenvalue greater than 1.0. This is because a component should at least explain as much variance as one single original variable.

- PC1 (2.525): Retain. It is nearly 2.5 times as powerful as any single variable.
- PC2 (1.245): Retain.
- PC3 (1.007): Retain (it just barely passes the threshold).

We can compute how much "information" each component retains by dividing their respective eigenvalues by the total variance of our dataset (8).

Component	**Eigenvalue**	**Variance(%)**	**Cumulative(%)**
PC1	2.525	31.6	31.6
PC2	1.245	15.6	47.2
PC3	1.007	12.6	59.8
PC4	0.978	12.2	72

Actuarial Interpretation:

- PC1 accounts for 31.6% of all the patterns in our portfolio. As seen in the correlation matrix above, this is the "Claims/History" engine. It dominates the analysis because claims and discounts are the highly volatile and interconnected parts of the data.
- There is a significant fall from PC1 to PC2, and another one from PC2 to PC3; known as "The Elbow". This confirms that the most important information is concentrated in the first 3 components.

2. Scree Plot

A scree plot further strengthens the arguments made above. The scree plot is a graphical method where we plot the eigenvalues, that is the variance, for each component and then we carry out the 'Elbow Test' where we look for a specific point where the slope of the graph shifts from a steep drop to a shallow tail. We can observe that there is a sharp fall from PC1 to PC2 and that the distinct elbow occurs at PC3. This suggests that the first three components are the most meaningful. Retaining further components provides diminishing returns.

This is illustrated in Fig 1.3 below:

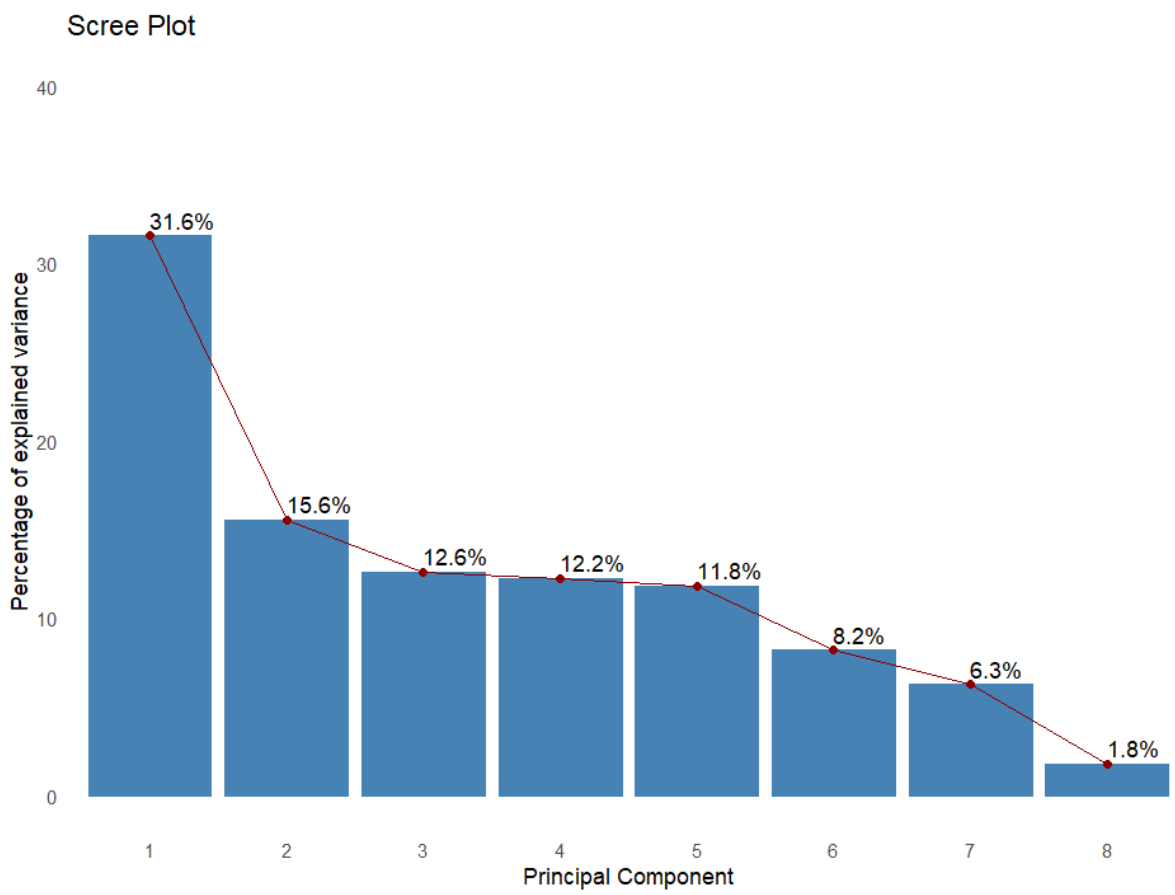

Fig1.3: Scree Plot

Ultimately, it can be observed that PC1 has the highest proportion of about 31.6%, followed by PC2 with 15.6% and finally PC3 with 12.6%. By using just 3 components instead of the original 8 variables, we are still retaining approximately 60% of the portfolio's total risk information.

Loadings

Now that we have selected the optimal principal components, we must understand how the variables in the dataset contribute to each principal component. A loading, in fact, represents the correlation between the original variable and a Principal Component. The table below summarises the loadings of PC1, PC2 and PC3.

Variable	PC1	PC2	PC3
Age	0.025	-0.205	-0.644
Vehicle age	0.021	0.160	-0.726
Premium	-0.326	0.502	0.033
No claims discount	0.497	0.085	0.018
Past claims count	-0.582	-0.113	-0.020
Past total paid	-0.550	-0.142	-0.016
Annual mileage	-0.055	0.507	-0.209
Engine power kW	0.011	0.619	0.110

For a sound interpretation of the loadings, we will apply Cohen Standard to deduce the strength of the relationship between the variable and the component. The ranges are given below:

- ✓ 0.10 to 0.29 indicates a weak relationship ; negligible influence on component
- ✓ 0.30 to 0.49 indicates a moderate relationship ; significant contributor
- ✓ 0.50 and above indicates a strong relationship ; variable is a primary driver

Through the application of PCA, we have reduced 8 complex variables into three insightful dimensions which will allow more effective assessment of risk we will now deal with targeted risk segmentation.

The next section will explore the interpretation and actuarial meaning of all 3 loadings selected.

Loadings of Component 1 (Claims Experience Factor)

Past claim counts (-0.582): This variable has a strong inverse relation with PC1, implying that as Past claims count increases, the PC1 score decreases.

Past total paid (-0.551): This variable has a strong negative relationship with PC1, showing that as Past total paid rises, the PC1 score falls.

No claims discount (0.497): This variable has a moderate positive relation with the component. This means that higher values of this variable are associated with slightly higher values of component 1.

From above, we can see that PC1 is dealing with historical loss performance of the policyholders**.** Since Past Claims Count and Past Total Paid share negative loadings with PC1, it illustrates that the rise in frequency and severity of past losses leads to a fall in the score of PC1. On the other hand, policyholders with high NCD levels shift toward the positive end. The inverse relationship is actuarially meaningful as it uses the mechanics of a Bonus-Malus System which is widely used in motor insurance to promote safe driving. In real world context, Mauritian insurers like SWAN implement this system as it enables the distinction between profitable and loss-making segments, providing a robust metric for risk-based pricing and underwriting enhancements.

The Malus Side (Negative PC1): High frequency claimants are clustered on the negative side of the axis and are considered as 'High-risk' clients

The Bonus Side (Positive PC1): Safe drivers are pushed to the positive side of the axis and are referred to as 'Low-risk' clients

This allows the actuarial team to effectively spot frequent claimants as a segment that requires higher risk premiums and safe drivers in turn possess high NCDs for having minimal claim counts. The figure below provides a graphical representation of the major contributors to PC1.

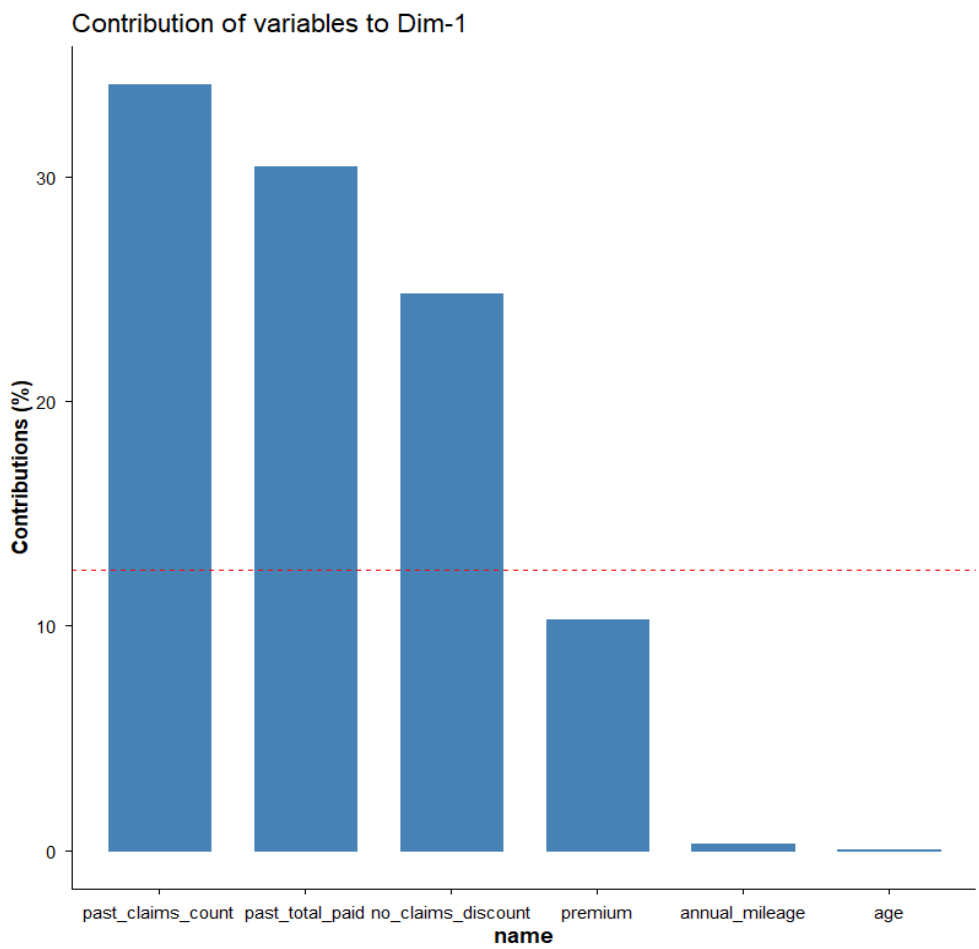

Fig1.5:PC1 Loadings

Loadings of Component 2 (Asset Exposure Factor)

Engine power kW (-0.619) : This variable has a strong negative loading on the second component meaning that they move in opposite directions.

Annual Mileage (-0.507) : This variable also contributes negatively but slightly as compared to Engine power kW.

Premium (-0.502) : Also considered as a primary driver of the second component since loading exceeds 0.50.

From the above variables, it is evident that we are now looking at *what* is being driven and *how* it is used. All the three variables above are strongly negative which is indicative of the fact that as a car becomes more powerful (Engine power), the more often it is driven (Annual Mileage) which in turn boosts risk exposure and leads to higher premium. As shown in the contribution analysis, the variables do not operate individually, rather, they form a cohesive profile of a 'High-Exposure Asset'. From an actuarial perspective, this means that the portfolio's risk is essentially structured around vehicle capability and usage intensity, validating that the current premium is highly sensitive to the physical properties of the insured asset.

Loadings of Component 3 (Life Stage and Asset Quality)

Vehicle age (-0.644): This variable has a strong negative loading on the third component, implying that as vehicle age decreases, the PC3 score increases.

Age (-0.726): Age has a stronger negative loading on PC3, making the relationship between PC3 and age an inverse one.

It can be observed that the loadings for age of drivers and the vehicle age are both strongly negative. This is a situation where we are looking at the age of the driver associated with the age of the car; a high PC3 score will indicate an older driver with an older vehicle whereas a low PC3 score will indicate a younger driver, likely with a newer vehicle.

PCA Biplots

To visualize the underlying structure of our motor insurance portfolios, PCA Biplots were generated. These biplots are essentially the "Risk Map", illustrating the 8-dimensional dataset into a 2D map. The first plot as shown below is defined by PC1 and PC2. This plot illustrates the loadings of each variable, where the direction and length of the arrows indicate their contribution to the model.

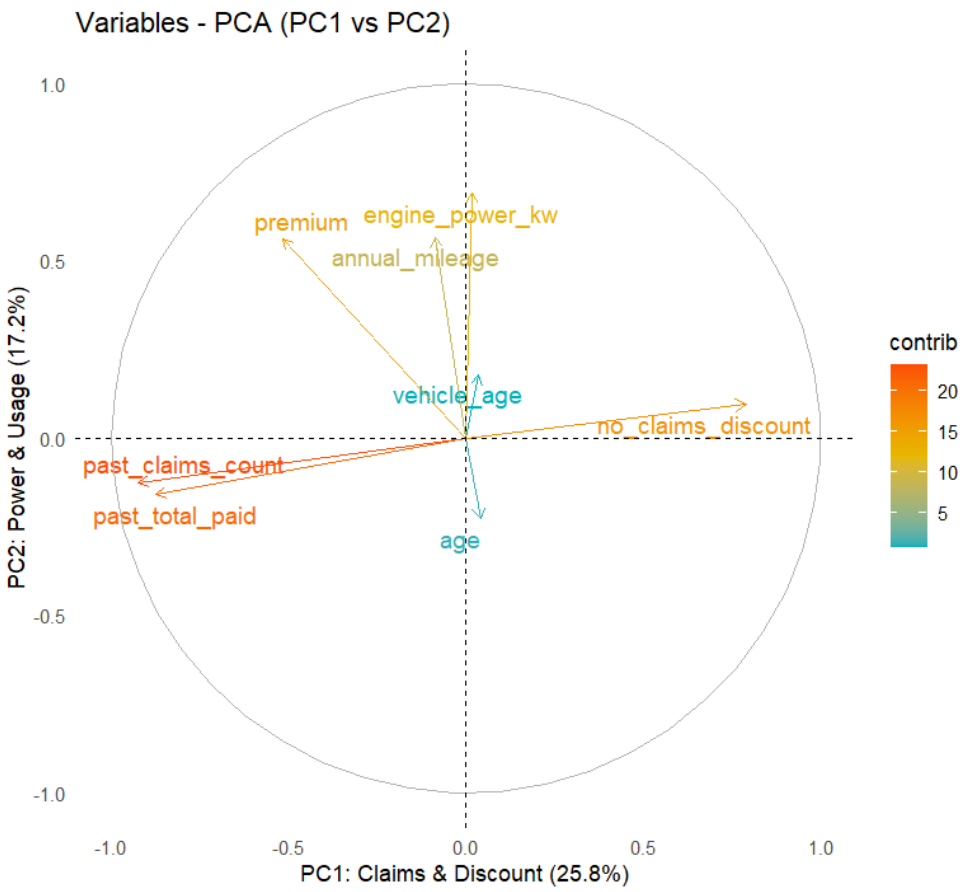

Fig 1.6: PC1 vs. PC2

Variable	PC1 Loading (Horizontal)	PC2 Loading (Vertical)	Actuarial Meaning
Past Claims Count	High Positive	Near Zero	Strongly defines the risky side of the Bonus-Malus axis.
Past Total Paid	High	Near Zero	Moves with claims, represents the financial severity of the risk.
No Claims Discount	High Negative	Near Zero	On the safe side and it is the direct opposite of claims.
Engine Power (kW)	Near Zero	High Positive	Only takes the car's potential into consideration and discards history.
Annual Mileage	Low Positive	High Positive	Represents "Exposure". High mileage pushes a profile upward into the danger zone.
Age/Vehicle Age	Low Negative	Low Negative	These pull towards the safe/standard zone, showing that maturity generally reduces accidents.

The second biplot is defined by PC1 and PC3. While the previous biplot focused on the car's power, this one shifts focus to the driver's maturity. It is essentially illustrating the age-related patterns.

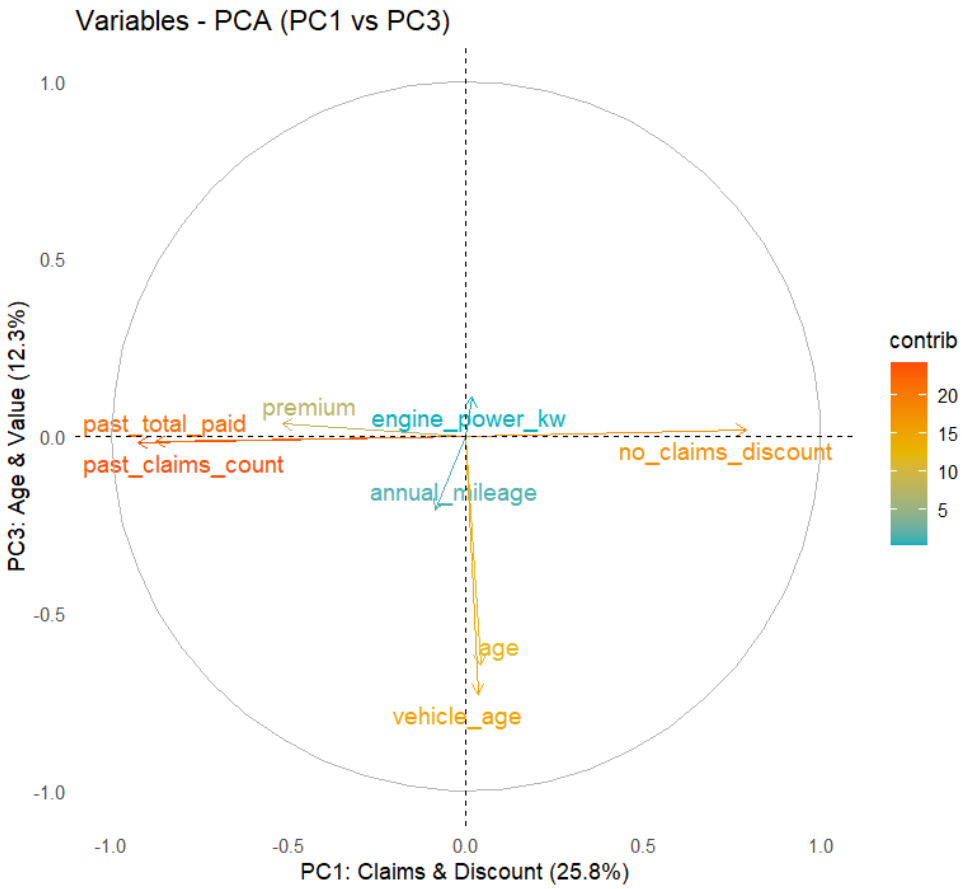

Fig 1.7: PC1 vs. PC3

Variable	PC1 Loading (Horizontal)	PC3 Loading (Vertical)	Actuarial Meaning
Age	Low Negative	High Positive	Primary driver of PC3. It shows that maturity is a major factor, independent of claims history.
Vehicle Age	Near Zero	High Positive	Closely correlated with driver age; older drivers tend to own older vehicles.
Past Claims Count	High Positive	Near Zero	It proves that claims happen across all age groups.
No Claims Discount	High Negative	Low Positive	This suggests that the best discounts are concentrated among older, safer drivers.
Premium	Low Positive	High Negative	This indicates that newer vehicles generally have higher premiums due to higher sum insured.

Robustness check

Now that we see these clear patterns, we can test for its robustness by removing variable 'Premium'. The same steps are carried out to perform PCA but this time without premium and the following results are obtained:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	1.5328	1.0392	1.0031	0.9826	0.9703	0.71528	0.38152
Proportion of Variance	0.3356	0.1543	0.1437	0.1379	0.1345	0.07309	0.02079
Cumulative Proportion	0.3356	0.4899	0.6337	0.7716	0.9061	0.097921	1.0000

It can be found that PC1 has the highest proportion of variance, of 33.56%, followed by PC2 with 15.43% and PC3 with 14.37%. The cumulative variance without premium is 63.36% which is only a 3.66% increase compared to the cumulative variance with premium (59.7%). This implies that the component structure remains largely stable without premium. PC1 still captures claim history vs. discount, PC2 still represents vehicle power and usage and PC3 still represents driver/vehicle age factors.

From the above analysis, there is consistency in the results even though the variable 'Premium' has been removed. This shows that the structure is robust and reliable as they are dependent on physical and behavioral data of policyholders rather than the pricing structure.

Emerging Patterns and Risk Profiles

By applying 75th-percentile thresholds to Principal Component scores, the portfolio was segmented into five distinct risk profiles. This classification allows for targeted underwriting interventions, specifically identifying the 'High Claims & High Power' segment which represents the highest combined frequency and severity risk to the motor insurance fund.

Component	Threshold	Description
PC1 (1.275)	**The Claims Threshold**	Anyone above **1.275** is in the top 25% for high claims, high payouts, and low discounts; representing our most expensive historical risks.
PC2 (0.728)	**The Exposure Threshold**	Anyone above **0.728** is in the top 25% for high engine power and high annual mileage; representing our high-usage drivers.
PC3 (0.712)	**The Maturity Threshold**	Anyone above **0.712** is in the top 25% for the old driver-old vehicle group.

The risk category distribution is as follows:

Risk Category	n	Percentage
Age-Related Risk	158	13.2
High Claims	263	21.9
High Claims & High Power	37	3.10
High Power/Usage	263	21.9
Standard Risk	479	39.9

The table reveals the difference in scales and distributions across the 1200 observations of the motor insurance portfolio and can be visualised through a PCA Score Plot as shown below:

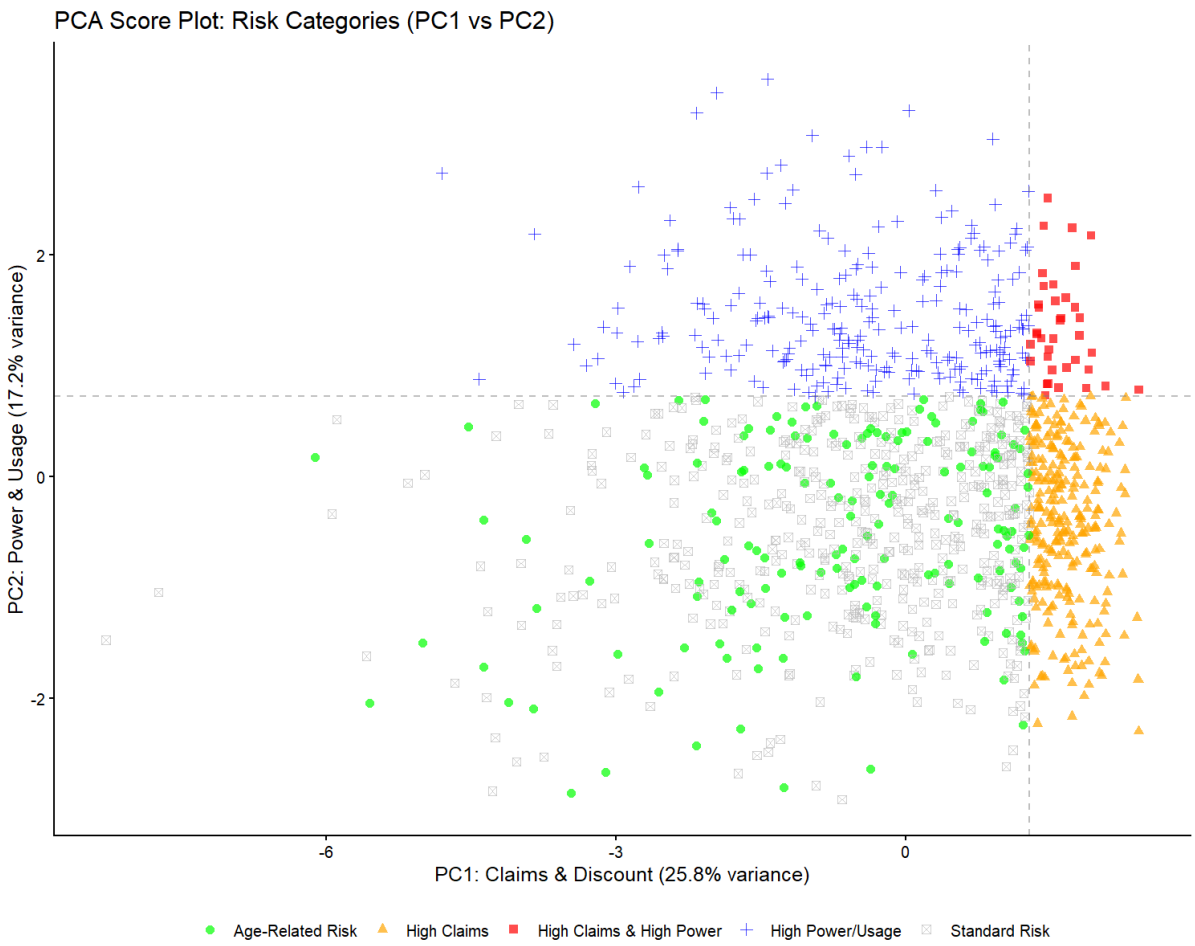

Fig 1.8: PCA Score Plot

The plot proves that the customers are not all the same; they naturally cluster into different zones based on their behaviour. The data is dense on the left side and thins out as you move right. This reflects the summary statistics—most people are "Safe" (left), and "High Claimants" (right) are the statistical outliers.

The variables like past_claims_count and past_total_paid point toward the right, confirming that customers in that area are the ones driving those numbers up.

The integration of PCA scoring allows for a clear visualization of the portfolio's riskiness. While the majority of the 1,200 policies are classified as 'Standard Risk,' the analysis successfully isolated approximately 30% of the portfolio into specialized risk pools, allowing the insurer to move away from a 'one-size-fits-all' pricing model toward a more risk-based pricing strategy.

Analysis of policies with recent claims:

- Number of policies with recent claims: 108
- Percentage of total portfolio: 9 %

Distinct Risk Profiles

After examining the subset of policyholders who have non-missing claim severity, the following 4 profiles with high risk were identified:

PROFILE 1: YOUNG POWER DRIVERS

Definition: Age < 30, Engine Power > 100 kW, at least 1 past claim.

Count	22
Average age	26.2 years
Average engine power	113.7 kW
Average premium	MUR 6459
Average past claims	1.2
Recent claim rate	4.5%

From the information provided above, it can be observed that for profile 1, the drivers are mostly below 30 years of age, who have crashed before and they pay the highest premiums because they are statistically the most dangerous combination. However, their recent claim rate is low, implying that the high premiums might be doing their job of cautioning them.

PROFILE 2: HIGH-MILEAGE COMMERCIAL USERS

Definition: Annual Mileage > 20,000 km/year, Diesel fuel type

Count	54
Average annual mileage	22 334 km/year
Average age	42.1 years
Average vehicle age	7 years
Average premium	MUR 5804
Average past claims	0.7
Recent claim rate	14.8%

Policyholders in profile 2 are individuals driving diesel cars over 20,000 km a year. Since they are on the road so much, they have a high chance of a recent claim (14.8%). This is because the more one drives, the more likely a mishap can occur.

PROFILE 3: AGED RISK

Definition: Age > 60 years with recent claims

Count	5
Average age	64.2 years
Average vehicle age	8.1 years
Average premium	MUR 5506
Average past claims	1.8
Average claim severity	MUR 66 198

Profile 3 is made up of drivers over 60 years with recent claims. There are very few of them but when they crash, it is extremely expensive to recover their losses (Rs 66,198 per claim). They represent a low frequency-high cost risk.

PROFILE 4: LOW DISCOUNT HIGH CLAIM

Definition: No-claims discount (NCD) < 20%, At least 2 past claims

Count	71
Average NCD	12.5%
Average past claims	2.7
Average total paid	MUR 32 629
Average premium	MUR 5984
Recent claim rate	21.1%

It can be deduced that policyholders are drivers with very low discounts and a history of multiple accidents. This pool has the highest count and the highest recent claim rate, therefore, concluding that Profile 4 has the most significant “Volume Risk”.

KEY FINDINGS FROM PCA ANALYSIS:

1. Three principal components explain 55.3% of total variance.

2. Component 1 (25.8%): Claims history vs. no-claims discount.

3. Component 2 (17.2%): Vehicle power and usage patterns.

4. Component 3 (12.3%): Driver and vehicle age factors.

ACTUARIAL IMPLICATIONS:

1. Premium differentiation should consider these risk dimensions.
2. Underwriting guidelines should address specific risk profiles.
3. Claims prevention programs could target high-risk groups.
4. Portfolio monitoring should track evolution of risk profiles.

Problem 2 – Association Between Categorical Variables

In the dataset provided there are four kinds of claim recorded and two transmission types. This study investigates whether claim_type and transmission type are independent categorical variables using the chi-square test of independence.

H_0 :There is an association between claim_type and transmission type of the vehicle.

H_1 :There is no association between claim_type and transmission type of the vehicle.

A one-tailed test is carried out at the 5% level of significance.

Assuming that the expected counts ≥ 5, under H_0 :

$\sum \frac{(O-E)^2}{E} \sim \chi^2_\alpha$, (r – 1)(c – 1),

Where,

O is the observed count, and

E is the expected count

Degrees of freedom, $\nu = (r-1)(c-1)$

= (4-1)(2-1)

= 3

Using R, the observed and expected counts were computed. Table 1 below summarizes the results with the respective expected counts in red font beside the observed ones.

Table 1: Contingency Table

	Automatic	Manual	**Row total**
Bodily Injury	94 (92.70)	112 (113.30)	**206**
Other	43 (51.75)	72 (63.25)	**115**
Property	253 (263.70)	333 (322.30)	**586**
Windscreen	150 (131.85)	143 (161.15)	**293**
Column total	**540**	**660**	**1200**

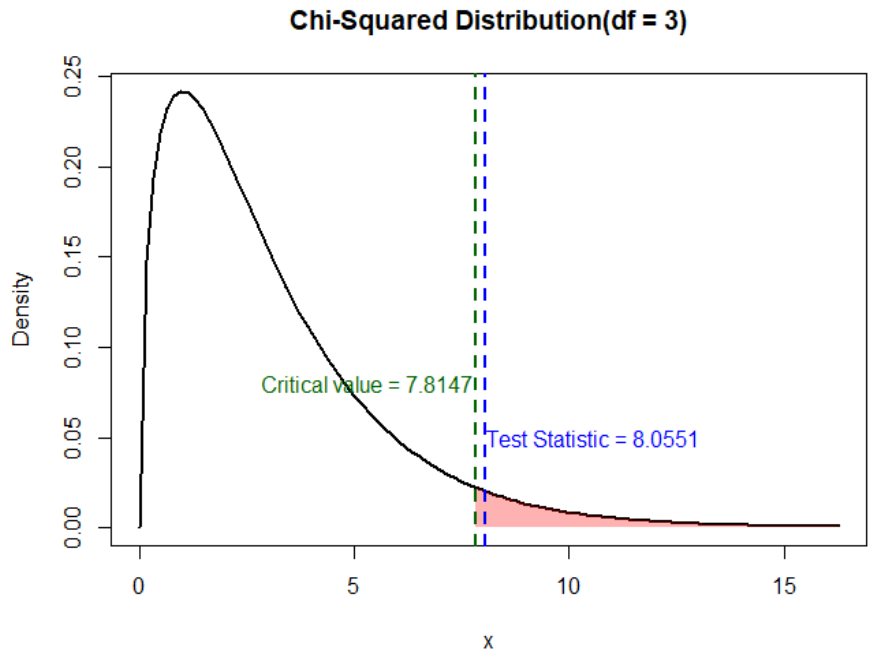

Fig 2.1

The chi-square test produced a test statistic of 8.0551 with 3 degrees of freedom and a p-value of 0.04489.

If chi-square test, $\chi^2$is greater than 7.8147, found in the shaded region of the graph in Fig 3.1 above, H_0 is rejected.

Since $\chi^2 = 8.0551 > 7.8147$ and the p-value is 0.04489 which is less than 0.05, we reject H_0 at the 5% level of significance and conclude that there is a relationship between claim type and transmission type.

Table 2: Standardised Residuals

	Automatic	Manual
Bodily Injury	0.200	-0.200
Other	-1.725	1.725
Property	-1.242	1.242
Windscreen	2.452	-2.452

Table 2 above contains the standardized residual values for each claim type. Most of the residual values fall within the range of -2 to 2, suggesting that their frequencies are close to what would be expected under independence except for Windscreen which has a residual value of 2.45 indicating that it occurs more frequently than expected and contributes mostly to the observed association between claim_type and transmission type.

Problem 3 - Assessing the fit of the Poisson distribution for the claim count

A chi-squared goodness-of-fit test for a Poisson distribution determines if he observed count data fits a theoretical Poisson model.

The Poisson distribution is a discrete probability distribution that models the count of random independent events within a fixed interval of time or space, based on a constant average rate.

H_0: The observed claim counts follow a Poisson Distribution.

H_1: The observed claim counts do not follow Poisson Distribution.

Assuming that expected counts ≥ 5, under H_0

Test statistics: $\Sigma \frac{(0-E)^2}{E}$

Degrees of Freedom: m - 1 – 1, where m is the number of categories

The mean of number of claims has a value of 4.255 with a variance of 4.635. Since these values are reasonably close, the Poisson assumption is reasonable.

The histogram shows that most days have 2 or 3 claims daily and fewer days having higher number of claims. The mean lies to the right of the peak making the histogram positively skewed. However, even if this pattern suggests that the Poisson assumption can be made possible, the data show evidence of slight overdispersion where variance is greater than mean.

The QQ-plot shows that the daily claim-counts align reasonably well with the central quantiles with deviations at the tails, which is a bit more than the Poisson model expects. The higher claim-counts sit above the diagonal showing overdispersion.

The chi square statistics are dependent on how far the observed frequencies differ from the expected Poisson frequencies.

The Chi-Square goodness of fit test produced a statistic of 9.773, with degree of freedom(df) is 10 and the corresponding p-value was 0.461. There is a moderate deviation since the value of the statistics is close to the df.

Since the p-value (0.461) is much greater than 0.05, we fail to reject the Null hypothesis H_0.

```
cat("START OF PROBLEM 3 ANALYSIS\n")
START OF PROBLEM 3 ANALYSIS
>cat("==============================================================\n)
==============================================================
> View(data2)
> str(data2)
'data.frame':	365 obs. of  2 variables:
 $ day       : int  1 2 3 4 5 6 7 8 9 10 ...
 $ claim_count: int  5 5 8 5 8 2 6 3 2 3 ...
> #column Names
> names(data2)
[1] "day"         "claim_count"
> #Calculaing the mean and the variance
> mean(data2$claim_count)
[1] 4.254795
> var(data2$claim_count)
[1] 4.635451
> x <- na.omit(as.numeric(data2$claim_count))
> #PLOT 9: QQ-plot Observed vs. Poisson
> pois_mean <- mean(x)
> theoretical <- qpois(ppoints(length(x)), lambda = pois_mean)
> qqplot(theoretical, sort(x),
+       main = "QQ-plot: Observed vs Poisson",
+       xlab = "Theoretical Quantiles (Poisson)",
+       ylab = "Observed Quantiles",
+       pch = 1, col="red")
> abline(0,1, col="red", lwd=2)
```

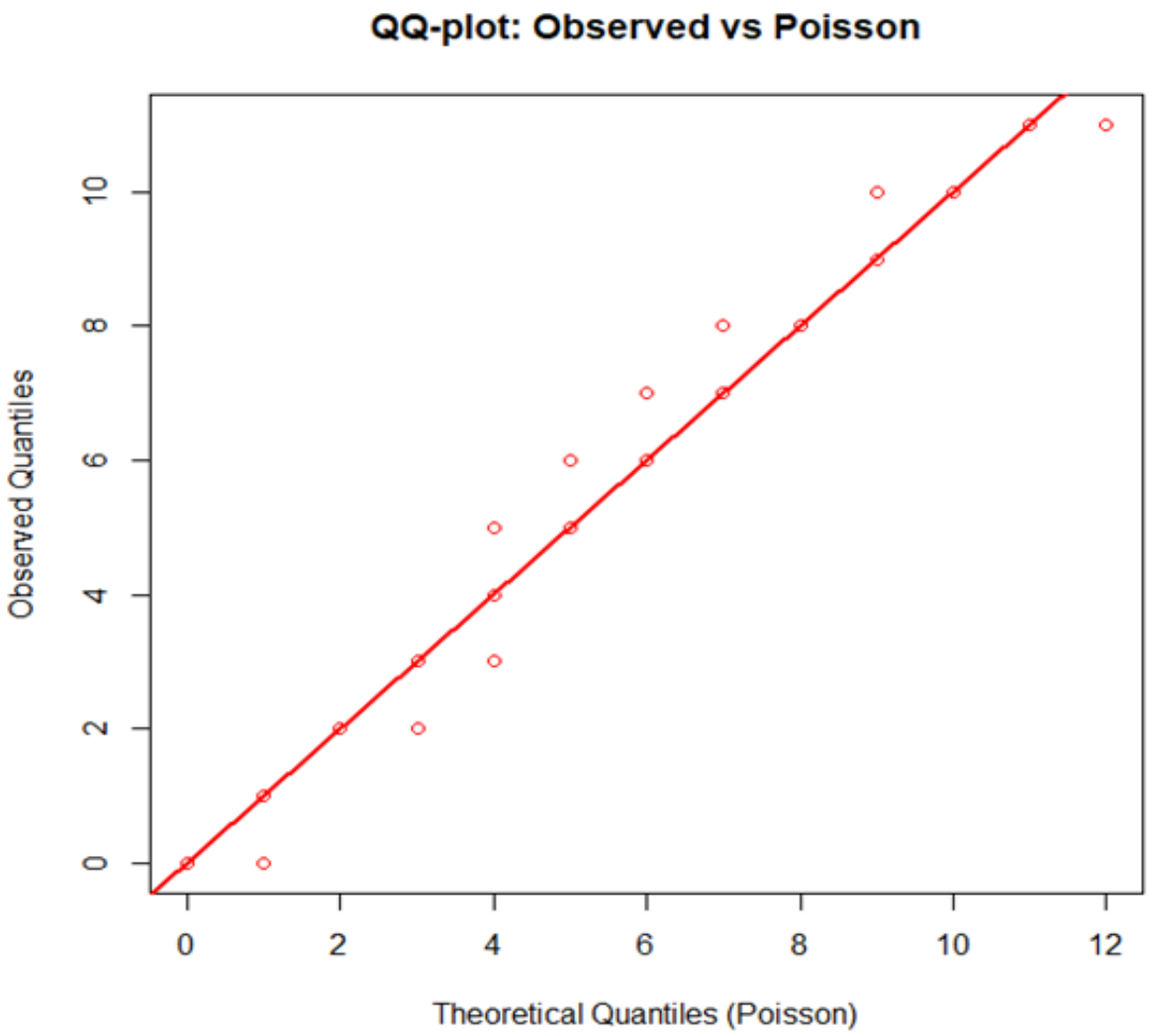

Fig 3.1

```
> #PLOT 10: Histogram of Number of Claims
> Number_Of_Claims<- data2$claim_count
> hist(Number_Of_Claims,
+     main = "Histogram of Number of Claims",
+     xlim = c(0,12),
+     ylim = c(0,70),
+     col = "#FFD1DC")
```

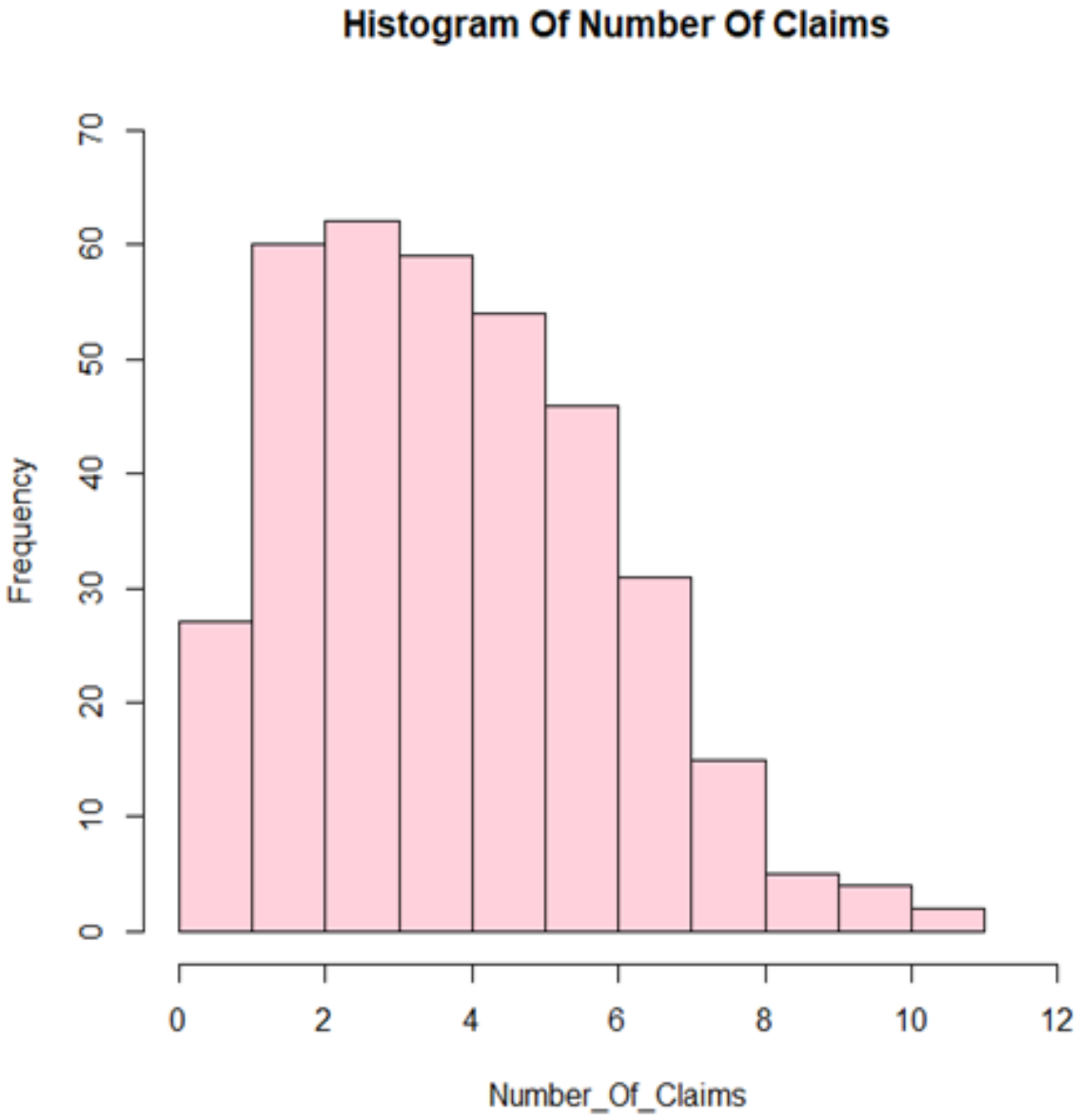

Fig 3.2

```
> #calculations for the chi-square goodness of fit test
> observed <- table(x)
> n <-  sum(observed)
> categories <- as.numeric(names(observed))
> expected_prob <- dpois(categories, lambda=pois_mean)
> expected <- expected_prob * n
> chi_sq <- sum((observed-expected)^2 / expected)
> df <- length(observed) - 1 - 1
```

```
> p_value <- 1 - pchisq(chi_sq, df)
>
> cat("Chi- Square Statistic:", chi_sq, "\n")
Chi- Square Statistic: 9.773264
>
> cat("Degree Of Freedom:", df, "\n")
Degree Of Freedom: 10
>
> cat("P-value:", p_value, "\n")
P-value: 0.4606059
>
> #End of Task 3
>
cat("\n==================================================================\n
")
==================================================================
> cat("END OF PROBLEM 3 ANALYSIS\n")
END OF PROBLEM 3 ANALYSIS
>
cat("==================================================================\n")
==================================================================
```

Word count: 3490

APPENDIX A

- Input in black, output in blue and comments in green

```
#Student ID:2519242
> cat("Seed:", 519242, "\n")
Seed: 519242
> #loading datasets
> data1 <- read.csv("umu_motor_portfolio_2025.csv", header = TRUE)
> data2 <- read.csv("umu_daily_claim_counts_2025.csv", header = TRUE)
> cat("TOKEN-A:", sum(data1$policy_id * (data1$past_claims_count + 1)) %% 100000,
"\n")
TOKEN-A: 44371
> cat("TOKEN-B:", sum(data2$claim_count[data2$day %in% c(7,77,177)]) * 100 +
sum(data2$claim_count == 0), "\n")
TOKEN-B: 2406
> # PROBLEM 1: RISK PROFILING AND DIMENSION REDUCTION
> # File: umu_motor_portfolio_2025.csv
>cat("\n==============================================================\
n")
==============================================================
> cat("START OF PROBLEM 1 ANALYSIS\n")
START OF PROBLEM 1 ANALYSIS
>cat("==============================================================\n"
)
==============================================================
> View(data1)
> str(data1)
'data.frame':	1200 obs. of  14 variables:
 $ policy_id        : int  1 2 3 4 5 6 7 8 9 10 ...
 $ age              : num  38 28 45 42 56 45 34 26 48 39 ...
 $ vehicle_age      : num  1.4 6 4.7 4.9 5.8 8.2 5 6.8 1.7 3.4 ...
 $ premium          : num  6462 6846 4496 5001 4000 ...
```

```
 $ no_claims_discount: num  44.5 52.1 29.8 41.9 54 41.7 40.4 43.2 33.3 29.2 ...
 $ past_claims_count : int  0 1 0 0 0 0 1 0 0 1 ...
 $ past_total_paid   : num  0 23064 0 0 0 ...
 $ annual_mileage    : num  24002 18130 19146 15944 12414 ...
 $ engine_power_kw   : num  82.1 104.8 75.5 NA 122.3 ...
 $ region            : chr  "North" "West" "West" "Central" ...
 $ fuel_type         : chr  "Petrol" "Petrol" "Hybrid" "Petrol" ...
 $ transmission      : chr  "Manual" "Manual" "Manual" "Manual" ...
 $ claim_type        : chr  "Windscreen" "Windscreen" "BodilyInjury" "Property" ...
 $ claim_severity    : num  NA NA NA NA NA NA NA NA NA NA ...
> names(data1)
 [1] "policy_id"         "age"               "vehicle_age"       "premium"
 [5] "no_claims_discount" "past_claims_count"  "past_total_paid"    "annual_mileage"
 [9] "engine_power_kw"    "region"            "fuel_type"         "transmission"
[13] "claim_type"         "claim_severity"
>
> #Installing Packages
> #Load required libraries
> install.packages("tidyverse")
> library(tidyverse)
> install.packages("ggplot2")
> library(ggplot2)
> install.packages("FactoMineR")
> library(FactoMineR)
> install.packages("factoextra")
> library(factoextra)
> install.packages("corrplot")
> library(corrplot)
> install.packages("gridExtra")
> library(gridExtra)
```

```
> install.packages("psych")
> library(psych)
> install.packages("naniar")
> library(naniar)
> install.packages("dplyr")
> library(dplyr)
> install.packages("ggpubr")
> library(ggpubr)
> install.packages("nFactors")
> library(nFactors)
# Display summary of data
> summary(data1)
policy_id          age          vehicle_age        premium        no_claims_discount
past_claims_count
 Min.   :   1.0   Min.   :18.00   Min.   : 0.200   Min.   : 4000   Min.   : 0.00      Min.   :0.0000
 1st Qu.: 300.8   1st Qu.:36.00   1st Qu.: 2.900   1st Qu.: 4000   1st Qu.:28.18      1st
Qu.:0.0000
 Median : 600.5   Median :43.00   Median : 5.200   Median : 5128   Median :35.30      Median
:1.0000
 Mean   : 600.5   Mean   :42.64   Mean   : 6.048   Mean   : 5366   Mean   :34.64      Mean
:0.7233
 3rd Qu.: 900.2   3rd Qu.:50.00   3rd Qu.: 8.200   3rd Qu.: 6277   3rd Qu.:41.92      3rd
Qu.:1.0000
 Max.   :1200.0   Max.   :78.00   Max.   :20.000   Max.   :10903   Max.   :60.00      Max.
:5.0000

past_total_paid annual_mileage  engine_power_kw     region          fuel_type
 Min.   :    0   Min.   : 5000   Min.   : 40.00   Length:1200        Length:1200
 1st Qu.:    0   1st Qu.:13427   1st Qu.: 72.40   Class :character   Class :character
 Median : 2462   Median :16143   Median : 89.90   Mode  :character   Mode  :character
 Mean   : 8864   Mean   :16266   Mean   : 90.24
 3rd Qu.:14138   3rd Qu.:19049   3rd Qu.:106.90
```

```
 Max.   :95666   Max.   :31330   Max.   :178.80
                 NA's   :33      NA's   :15
 transmission       claim_type       claim_severity
 Length:1200        Length:1200        Min.   :   234.1
 Class :character   Class :character   1st Qu.:  2381.5
 Mode  :character   Mode  :character   Median : 19184.1
                                        Mean   : 53343.9
                                        3rd Qu.: 84604.7
                                        Max.   :324794.7
                                        NA's   :1092
> # Check for missing values
> na.counts<-colSums(is.na(data1))
> print(na.counts)
       policy_id             age       vehicle_age           premium no_claims_discount
               0               0               0                 0                  0
 past_claims_count   past_total_paid    annual_mileage   engine_power_kw            region
               0               0              33                15                  0
       fuel_type      transmission        claim_type     claim_severity
               0               0               0              1092
>
> pca_vars <- c("age", "vehicle_age", "premium",
"no_claims_discount","past_claims_count", "past_total_paid", "annual_mileage",
"engine_power_kw")
> cat("Variables selected for PCA:\n")
Variables selected for PCA:
> cat(paste(pca_vars, collapse = ", "), "\n\n")
age, vehicle_age, premium, no_claims_discount, past_claims_count, past_total_paid,
annual_mileage, engine_power_kw
> # Create dataset for PCA
> pca_data <- data1 %>%
+   select(all_of(pca_vars))
```

```
> # Impute missing values
> pca_data_imputed <- pca_data
> imputation_summary <- data.frame()
> for (var in pca_vars) {
+   missing_count <- sum(is.na(pca_data_imputed[[var]]))
+   if (missing_count > 0) {
+     median_val <- median(pca_data_imputed[[var]], na.rm = TRUE)
+     pca_data_imputed[[var]][is.na(pca_data_imputed[[var]])] <- median_val
+     imputation_summary <- rbind(imputation_summary,
+                                 data.frame(Variable = var,
+                                            Missing_Count = missing_count,
+                                            Imputation_Value = median_val,
+                                            Imputation_Type = "Median"))
+   }
+ }
> if (nrow(imputation_summary) > 0) {
+   cat("\nIMPUTATION SUMMARY:\n")
+   print(imputation_summary)
+ } else {
+   cat("\nNo missing values in selected variables.\n")
+ }

IMPUTATION SUMMARY:
        Variable Missing_Count Imputation_Value Imputation_Type
1  annual_mileage          33         16143.0         Median
2 engine_power_kw          15            89.9         Median
>
> # Verify no missing values remain
> cat("\nMissing values after imputation:", sum(is.na(pca_data_imputed)), "\n")
```

```
Missing values after imputation: 0
>
> #Plot 1: Boxplots for visual reprensentation of outliers
> par(mfrow = c(3, 3), mar = c(4, 4, 3, 1))
> boxplot(data1$age, main="Age", xlab="Years", col="lightblue", horizontal=TRUE)
> boxplot(data1$vehicle_age, main="Vehicle Age", xlab="Years", col="lightgreen",
horizontal=TRUE)
> boxplot(data1$premium, main="Premium", xlab="MUR", col="lightpink",
horizontal=TRUE)
> boxplot(data1$no_claims_discount, main="NCD", xlab="%", col="yellow",
horizontal=TRUE)
> boxplot(data1$past_claims_count, main="Past Claims", xlab="Count", col="navajowhite",
horizontal=TRUE)
> boxplot(data1$past_total_paid, main="Past Total Paid", xlab="MUR", col="plum",
horizontal=TRUE)
> boxplot(data1$annual_mileage, main="Annual Mileage", xlab="km/year", col="blue",
horizontal=TRUE)
> boxplot(data1$claim_severity, main="Claim Severity", xlab="MUR/Claim", col="salmon",
horizontal=TRUE)
> boxplot(data1$engine_power_kw, main="Engine power", xlab="kW", col="firebrick",
horizontal=TRUE)
>
> # Standardize the data
> pca_data_scaled <- scale(pca_data_imputed)
> summary(pca_data_scaled)
     age          vehicle_age        premium        no_claims_discount past_claims_count
 Min.   :-2.38580   Min.   :-1.4112   Min.   :-1.0188   Min.   :-3.31441   Min.   :-0.8395
 1st Qu.:-0.64293   1st Qu.:-0.7597   1st Qu.:-1.0188   1st Qu.:-0.61821   1st Qu.:-0.8395
 Median : 0.03486   Median :-0.2047   Median :-0.1777   Median : 0.06361   Median : 0.3211
 Mean   : 0.00000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.00000   Mean   : 0.0000
 3rd Qu.: 0.71264   3rd Qu.: 0.5192   3rd Qu.: 0.6797   3rd Qu.: 0.69759   3rd Qu.: 0.3211
 Max.   : 3.42378   Max.   : 3.3665   Max.   : 4.1308   Max.   : 2.42727   Max.   : 4.9635
 past_total_paid   annual_mileage     engine_power_kw
```

```
 Min.   :-0.6948   Min.   :-2.79342   Min.   :-2.06626
 1st Qu.:-0.6948   1st Qu.:-0.68407   1st Qu.:-0.71909
 Median :-0.5018   Median :-0.02972   Median :-0.01363
 Mean   : 0.0000   Mean   : 0.00000   Mean   : 0.00000
 3rd Qu.: 0.4134   3rd Qu.: 0.66523   3rd Qu.: 0.68155
 Max.   : 6.8046   Max.   : 3.73697   Max.   : 3.64327
>
> #Plot 2: Correlation Heatmap
> corr_matrix<- cor(pca_data_scaled)
> par(mfrow = c(1, 1))
> corrplot(corr_matrix, method = "color", type = "upper",tl.col = "black", tl.cex = 0.8, title =
"Correlation Heatmap")
>
> # Matrix of correlated data
> round((corr_matrix),3)
                    age vehicle_age premium no_claims_discount past_claims_count
past_total_paid
age               1.000       0.004  -0.055            0.001          -0.022         -0.012
vehicle_age       0.004       1.000   0.004            0.027          -0.015         -0.022
premium          -0.055       0.004   1.000           -0.303           0.330          0.264
no_claims_discount  0.001     0.027  -0.303            1.000          -0.631         -0.526
past_claims_count  -0.022     -0.015   0.330           -0.631           1.000          0.842
past_total_paid    -0.012     -0.022   0.264           -0.526           0.842          1.000
annual_mileage     -0.016      0.030   0.179           -0.020           0.024          0.005
engine_power_kw    -0.032      0.025   0.195            0.053          -0.055         -0.046
                 annual_mileage engine_power_kw
age                    -0.016         -0.032
vehicle_age             0.030          0.025
premium                 0.179          0.195
no_claims_discount     -0.020          0.053
past_claims_count       0.024         -0.055
```

```
past_total_paid          0.005        -0.046
annual_mileage           1.000         0.050
engine_power_kw          0.050         1.000
>
> #PRINCIPAL COMPONENT ANALYSIS
> cat("\n\nPRINCIPAL COMPONENT ANALYSIS\n")

PRINCIPAL COMPONENT ANALYSIS
>
> # Perform PCA
> pca_result <- prcomp(pca_data_scaled, center = TRUE, scale. = TRUE)
>
> # Summary of PCA
> cat("PCA SUMMARY:\n")
PCA SUMMARY:
> print(summary(pca_result))
Importance of components:
                          PC1    PC2    PC3    PC4    PC5     PC6     PC7     PC8
Standard deviation     1.5889 1.1156 1.0036 0.9891 0.9718 0.80960 0.70793 0.37986
Proportion of Variance 0.3156 0.1556 0.1259 0.1223 0.1181 0.08193 0.06265 0.01804
Cumulative Proportion  0.3156 0.4711 0.5970 0.7193 0.8374 0.91932 0.98196 1.00000
>
> # Extract eigenvalues
> eigenvalues <- pca_result$sdev^2
> cat("\nEIGENVALUES:\n")

EIGENVALUES:
> for (i in 1:length(eigenvalues)) {
+   cat(sprintf("PC%d: %.3f\n", i, eigenvalues[i]))
+ }
```

```
PC1: 2.525
PC2: 1.245
PC3: 1.007
PC4: 0.978
PC5: 0.944
PC6: 0.655
PC7: 0.501
PC8: 0.144
>
> #Plot 3: Scree Plot for PCA analysis
> fviz_eig(pca_result,xlab="Principal Component",ylab="Percentage of explained variance",main="Scree Plot", addlabels = TRUE,ylim = c(0, 40), linecolor ="darkred") +
+   theme_minimal() +
+   theme(panel.grid.major = element_blank(),
+         panel.grid.minor = element_blank())
>
> # Cumulative variance
> cum_var <- cumsum(pca_result$sdev^2 / sum(pca_result$sdev^2))
> cat("\nCUMULATIVE VARIANCE EXPLAINED:\n")

CUMULATIVE VARIANCE EXPLAINED:
> for (i in 1:length(cum_var)) {
+   cat(sprintf("PC1 to PC%d: %.1f%%\n", i, cum_var[i] * 100))
+ }
PC1 to PC1: 31.6%
PC1 to PC2: 47.1%
PC1 to PC3: 59.7%
PC1 to PC4: 71.9%
PC1 to PC5: 83.7%
PC1 to PC6: 91.9%
PC1 to PC7: 98.2%
```

```
PC1 to PC8: 100.0%
>
> # Decide on number of components (Kaiser Criterion)
> n_components <- 3
> cat(sprintf("\nDECISION: Retaining %d principal components\n", n_components))

DECISION: Retaining 3 principal components
> cat(sprintf("Variance explained by first %d PCs: %.1f%%\n",
+            n_components, cum_var[n_components] * 100))
Variance explained by first 3 PCs: 59.7%
>
> #COMPONENT INTERPRETATION
> cat("\n\nCOMPONENT INTERPRETATION\n")

COMPONENT INTERPRETATION
>
> # Loadings for first 3 components
> loadings <- pca_result$rotation[, 1:n_components]
> colnames(loadings) <- paste0("PC", 1:n_components)
>
> cat("VARIABLE LOADINGS (First 3 Principal Components):\n")
VARIABLE LOADINGS (First 3 Principal Components):
> print(round(loadings, 3))
                   PC1    PC2    PC3
age                0.025 -0.205 -0.644
vehicle_age        0.021  0.160 -0.726
premium           -0.326  0.502  0.033
no_claims_discount  0.497  0.085  0.018
past_claims_count  -0.582 -0.113 -0.020
past_total_paid    -0.551 -0.142 -0.016
```

```
annual_mileage     -0.055  0.507 -0.209
engine_power_kw     0.011  0.619  0.110
> #Plot 4: Contribution of variable for PC1
> fviz_contrib(pca_result, choice = "var", axes = 1, top = 6)+
+   theme_minimal() + # Use minimal theme
+   theme(panel.grid.major = element_blank(), # Remove major grid lines
+         panel.grid.minor = element_blank(),, # Remove minor grid lines
+         axis.line = element_line(color = "black"), # Add X and Y axis lines
+         axis.ticks = element_line(color = "black"), # Add axis ticks
+         axis.text = element_text(color = "black"), # Ensure axis text is visible
+         axis.title = element_text(face = "bold")) # Make axis titles bold
> # PLOT 5: PC1 vs PC2 biplot
> biplot_1 <- fviz_pca_var(pca_result,
+                          axes = c(1, 2),
+                          col.var = "contrib",
+                          gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
+                          repel = TRUE,
+                          title = "Variables - PCA (PC1 vs PC2)") +
+   xlab("PC1: Claims & Discount (25.8%)") +
+   ylab("PC2: Power & Usage (17.2%)") +
+   theme_minimal() +
+   theme(panel.grid.major = element_blank(),
+         panel.grid.minor = element_blank())
> print(biplot_1)
> # PLOT 6: PC1 vs PC3 biplot
> biplot_2 <- fviz_pca_var(pca_result,
+                          axes = c(1, 3),
+                          col.var = "contrib",
+                          gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
+                          repel = TRUE,
```

```
+                    title = "Variables - PCA (PC1 vs PC3)") +
+   xlab("PC1: Claims & Discount (25.8%)") +
+   ylab("PC3: Age & Value (12.3%)") +
+   theme_minimal() +
+   theme(panel.grid.major = element_blank(),
+         panel.grid.minor = element_blank())
> print(biplot_2)
> #ROBUSTNESS CHECK (EXCLUDING PREMIUM)
> cat("\n\nROBUSTNESS CHECK (EXCLUDING PREMIUM)\n")
ROBUSTNESS CHECK (EXCLUDING PREMIUM)
> # Select variables excluding premium
> pca_vars_no_premium <- setdiff(pca_vars, "premium")
> cat("Variables for PCA without premium:\n")
Variables for PCA without premium:
> cat(paste(pca_vars_no_premium, collapse = ", "), "\n\n")
age, vehicle_age, no_claims_discount, past_claims_count, past_total_paid, annual_mileage,
engine_power_kw
> pca_data_no_premium <- data1 %>%
+   select(all_of(pca_vars_no_premium))
> # Impute missing values
> for (var in pca_vars_no_premium) {
+   if (any(is.na(pca_data_no_premium[[var]]))) {
+     median_val <- median(pca_data_no_premium[[var]], na.rm = TRUE)
+     pca_data_no_premium[[var]][is.na(pca_data_no_premium[[var]])] <- median_val
+   }
+ }
> # Standardize
> pca_data_no_premium_scaled <- scale(pca_data_no_premium)
>
> # Perform PCA without premium
> pca_result_no_premium <- prcomp(pca_data_no_premium_scaled,
```

```
+                                 center = TRUE, scale. = TRUE)
> cat("PCA SUMMARY (without premium):\n")
PCA SUMMARY (without premium):
> print(summary(pca_result_no_premium))
Importance of components:
                          PC1    PC2    PC3    PC4    PC5     PC6     PC7
Standard deviation     1.5328 1.0392 1.0031 0.9826 0.9703 0.71528 0.38152
Proportion of Variance 0.3356 0.1543 0.1437 0.1379 0.1345 0.07309 0.02079
Cumulative Proportion  0.3356 0.4899 0.6337 0.7716 0.9061 0.97921 1.00000
> # Compare variance explained
> cum_var_no_premium <- cumsum(pca_result_no_premium$sdev^2 /
+                                 sum(pca_result_no_premium$sdev^2))
> cat(sprintf("\nVARIANCE EXPLAINED COMPARISON (first 3 PCs):\n"))
VARIANCE EXPLAINED COMPARISON (first 3 PCs):
> cat(sprintf("With premium: %.1f%%\n", cum_var[3] * 100))
With premium: 59.7%
> cat(sprintf("Without premium: %.1f%%\n", cum_var_no_premium[3] * 100))
Without premium: 63.4%
> cat("\nCONCLUSION: Component structure remains largely stable without premium.\n")
CONCLUSION: Component structure remains largely stable without premium.
> cat("PC1 still captures claims history vs. discount\n")
PC1 still captures claims history vs. discount
> cat("PC2 still represents vehicle power and usage\n")
PC2 still represents vehicle power and usage
> cat("PC3 still represents driver/vehicle age factors\n")
PC3 still represents driver/vehicle age factors
> # EMERGING PATTERNS AND RISK PROFILES
> cat("\n\nEMERGING PATTERNS AND RISK PROFILES\n")
EMERGING PATTERNS AND RISK PROFILES
> # Calculate component scores
```

```
> scores <- as.data.frame(pca_result$x[, 1:3])
> names(scores) <- c("PC1", "PC2", "PC3")
> # Add scores to original data
> portfolio_with_scores <- cbind(data1, scores)
> # Identify high-risk clusters
> thresholds <- apply(scores, 2, quantile, probs = 0.75, na.rm = TRUE)
> cat("Thresholds for high scores (75th percentile):\n")
Thresholds for high scores (75th percentile):
> print(thresholds)
     PC1       PC2       PC3
1.2756864 0.7282641 0.7124546
> # Identify policies in each high-risk category
> portfolio_with_scores <- portfolio_with_scores %>%
+   mutate(
+     high_risk_PC1 = PC1 > thresholds["PC1"],
+     high_risk_PC2 = PC2 > thresholds["PC2"],
+     high_risk_PC3 = PC3 > thresholds["PC3"],
+     any_high_risk = high_risk_PC1 | high_risk_PC2 | high_risk_PC3,
+     risk_category = case_when(
+       high_risk_PC1 & high_risk_PC2 ~ "High Claims & High Power",
+       high_risk_PC1 ~ "High Claims",
+       high_risk_PC2 ~ "High Power/Usage",
+       high_risk_PC3 ~ "Age-Related Risk",
+       TRUE ~ "Standard Risk"
+     )
+   )
> # Count risk categories
> risk_counts <- portfolio_with_scores %>%
+   count(risk_category) %>%
+   mutate(percentage = round(n / sum(n) * 100, 1))
```

```
> cat("\nRISK CATEGORY DISTRIBUTION:\n")

RISK CATEGORY DISTRIBUTION:
> print(risk_counts)
           risk_category   n percentage
1       Age-Related Risk 158       13.2
2           High Claims 263       21.9
3 High Claims & High Power  37        3.1
4       High Power/Usage 263       21.9
5          Standard Risk 479       39.9
> # PLOT 7: Score plot with risk categories
> score_plot <- ggplot(portfolio_with_scores, aes(x = PC1, y = PC2)) +
+   geom_point(aes(color = risk_category, shape = risk_category), alpha = 0.7, size = 2) +
+   geom_hline(yintercept = thresholds["PC2"], linetype = "dashed",
+              color = "gray50", alpha = 0.5) +
+   geom_vline(xintercept = thresholds["PC1"], linetype = "dashed",
+              color = "gray50", alpha = 0.5) +
+   scale_color_manual(values = c("High Claims & High Power" = "red",
+                              "High Claims" = "orange",
+                              "High Power/Usage" = "blue",
+                              "Age-Related Risk" = "green",
+                              "Standard Risk" = "gray")) +
+   ggtitle("PCA Score Plot: Risk Categories (PC1 vs PC2)") +
+   xlab("PC1: Claims & Discount (25.8% variance)") +
+   ylab("PC2: Power & Usage (17.2% variance)") +
+   theme_minimal() +
    theme(panel.grid.major = element_blank(),
      panel.grid.minor = element_blank(),
      legend.position = "bottom",
      legend.title = element_blank())
```

```
> print(score_plot)
> # Analyze policies with recent claims
> recent_claims <- portfolio_with_scores %>%
+   dplyr::filter(!is.na(claim_severity))
> cat("\nANALYSIS OF POLICIES WITH RECENT CLAIMS:\n")
ANALYSIS OF POLICIES WITH RECENT CLAIMS:
> cat("Number of policies with recent claims:", nrow(recent_claims), "\n")
Number of policies with recent claims: 108
> cat("Percentage of total portfolio:",
+     round(nrow(recent_claims)/nrow(data1)*100, 1), "%\n")
Percentage of total portfolio: 9 %
> #DISTINCT RISK PROFILES
> cat("\n\nDISTINCT RISK PROFILES\n")
DISTINCT RISK PROFILES
> names(portfolio_with_scores)
 [1] "policy_id"         "age"               "vehicle_age"       "premium"
 [5] "no_claims_discount" "past_claims_count"  "past_total_paid"    "annual_mileage"
 [9] "engine_power_kw"    "region"            "fuel_type"         "transmission"
[13] "claim_type"         "claim_severity"     "PC1"               "PC2"
[17] "PC3"                "high_risk_PC1"      "high_risk_PC2"      "high_risk_PC3"
[21] "any_high_risk"      "risk_category"
> # Profile 1: Young Power Drivers
> young_power_drivers <- portfolio_with_scores %>%
+   dplyr::filter(age < 30,
+         engine_power_kw > 100,
+         past_claims_count >= 1)
> cat("PROFILE 1: YOUNG POWER DRIVERS\n")
PROFILE 1: YOUNG POWER DRIVERS
> cat("Definition: Age < 30, Engine Power > 100 kW, At least 1 past claim\n")
Definition: Age < 30, Engine Power > 100 kW, At least 1 past claim
```

```
> cat("Count:", nrow(young_power_drivers), "\n")
Count: 22
> if (nrow(young_power_drivers) > 0) {
+   young_power_summary <- young_power_drivers %>%
+     summarise(
+       avg_age = round(mean(age, na.rm = TRUE), 1),
+       avg_engine_power = round(mean(engine_power_kw, na.rm = TRUE), 1),
+       avg_premium = round(mean(premium, na.rm = TRUE), 0),
+       avg_past_claims = round(mean(past_claims_count, na.rm = TRUE), 1),
+       recent_claim_rate = round(sum(!is.na(claim_severity)) / n() * 100, 1)
+     )
+   cat("Average age:", young_power_summary$avg_age, "years\n")
+   cat("Average engine power:", young_power_summary$avg_engine_power, "kW\n")
+   cat("Average premium: MUR", young_power_summary$avg_premium, "\n")
+   cat("Average past claims:", young_power_summary$avg_past_claims, "\n")
+   cat("Recent claim rate:", young_power_summary$recent_claim_rate, "%\n")
+ }
Average age: 26.2 years
Average engine power: 113.7 kW
Average premium: MUR 6459
Average past claims: 1.2
Recent claim rate: 4.5 %
> cat("\n")
> # Profile 2: High-Mileage Commercial Users
> high_mileage_commercial <- portfolio_with_scores %>%
+   dplyr::filter(annual_mileage > 20000,
+         fuel_type == "Diesel")
> cat("PROFILE 2: HIGH-MILEAGE COMMERCIAL USERS\n")
PROFILE 2: HIGH-MILEAGE COMMERCIAL USERS
> cat("Definition: Annual Mileage > 20,000 km/year, Diesel fuel type\n")
```

```
Definition: Annual Mileage > 20,000 km/year, Diesel fuel type
> cat("Count:", nrow(high_mileage_commercial), "\n")
Count: 54
> if (nrow(high_mileage_commercial) > 0) {
+   high_mileage_summary <- high_mileage_commercial %>%
+     summarise(
+       avg_mileage = round(mean(annual_mileage, na.rm = TRUE), 0),
+       avg_age = round(mean(age, na.rm = TRUE), 1),
+       avg_vehicle_age = round(mean(vehicle_age, na.rm = TRUE), 1),
+       avg_premium = round(mean(premium, na.rm = TRUE), 0),
+       avg_past_claims = round(mean(past_claims_count, na.rm = TRUE), 1),
+       recent_claim_rate = round(sum(!is.na(claim_severity)) / n() * 100, 1)
+     )
+   cat("Average annual mileage:", high_mileage_summary$avg_mileage, "km/year\n")
+   cat("Average age:", high_mileage_summary$avg_age, "years\n")
+   cat("Average vehicle age:", high_mileage_summary$avg_vehicle_age, "years\n")
+   cat("Average premium: MUR", high_mileage_summary$avg_premium, "\n")
+   cat("Average past claims:", high_mileage_summary$avg_past_claims, "\n")
+   cat("Recent claim rate:", high_mileage_summary$recent_claim_rate, "%\n")
+ }
Average annual mileage: 22334 km/year
Average age: 42.1 years
Average vehicle age: 7 years
Average premium: MUR 5804
Average past claims: 0.7
Recent claim rate: 14.8 %
> cat("\n")
> # Profile 3: Aged Risk
> aged_risk <- portfolio_with_scores %>%
+   dplyr::filter(age > 60,
```

```
+         !is.na(claim_severity))
> cat("PROFILE 3: AGED RISK\n")
PROFILE 3: AGED RISK
> cat("Definition: Age > 60 years with recent claims\n")
Definition: Age > 60 years with recent claims
> cat("Count:", nrow(aged_risk), "\n")
Count: 5
> if (nrow(aged_risk) > 0) {
+   aged_risk_summary <- aged_risk %>%
+     summarise(
+       avg_age = round(mean(age, na.rm = TRUE), 1),
+       avg_vehicle_age = round(mean(vehicle_age, na.rm = TRUE), 1),
+       avg_premium = round(mean(premium, na.rm = TRUE), 0),
+       avg_past_claims = round(mean(past_claims_count, na.rm = TRUE), 1),
+       avg_claim_severity = round(mean(claim_severity, na.rm = TRUE), 0)
+     )
+   cat("Average age:", aged_risk_summary$avg_age, "years\n")
+   cat("Average vehicle age:", aged_risk_summary$avg_vehicle_age, "years\n")
+   cat("Average premium: MUR", aged_risk_summary$avg_premium, "\n")
+   cat("Average past claims:", aged_risk_summary$avg_past_claims, "\n")
+   cat("Average claim severity: MUR", aged_risk_summary$avg_claim_severity, "\n")
+ }
Average age: 64.2 years
Average vehicle age: 8.1 years
Average premium: MUR 5506
Average past claims: 1.8
Average claim severity: MUR 66198
> # Profile 4: Low Discount High Claim
> low_discount_high_claim <- portfolio_with_scores %>%
+   dplyr::filter(no_claims_discount < 20,
```

```
+         past_claims_count >= 2)
> cat("\nPROFILE 4: LOW DISCOUNT HIGH CLAIM\n")
PROFILE 4: LOW DISCOUNT HIGH CLAIM
> cat("Definition: No-claims discount < 20%, At least 2 past claims\n")
Definition: No-claims discount < 20%, At least 2 past claims
> cat("Count:", nrow(low_discount_high_claim), "\n")
Count: 71
> if (nrow(low_discount_high_claim) > 0) {
+   low_discount_summary <- low_discount_high_claim %>%
+     summarise(
+       avg_discount = round(mean(no_claims_discount, na.rm = TRUE), 1),
+       avg_past_claims = round(mean(past_claims_count, na.rm = TRUE), 1),
+       avg_total_paid = round(mean(past_total_paid, na.rm = TRUE), 0),
+       avg_premium = round(mean(premium, na.rm = TRUE), 0),
+       recent_claim_rate = round(sum(!is.na(claim_severity)) / n() * 100, 1)
+     )
+   cat("Average no-claims discount:", low_discount_summary$avg_discount, "%\n")
+   cat("Average past claims:", low_discount_summary$avg_past_claims, "\n")
+   cat("Average total paid: MUR", low_discount_summary$avg_total_paid, "\n")
+   cat("Average premium: MUR", low_discount_summary$avg_premium, "\n")
+   cat("Recent claim rate:", low_discount_summary$recent_claim_rate, "%\n")
+ }
Average no-claims discount: 12.5 %
Average past claims: 2.7
Average total paid: MUR 32629
Average premium: MUR 5984
Recent claim rate: 21.1 %
> #SUMMARY AND CONCLUSIONS
> cat("\n\nSUMMARY AND CONCLUSIONS\n")
SUMMARY AND CONCLUSIONS
```

```
> cat("==================================================\n\n")
==================================================
> cat("KEY FINDINGS FROM PCA ANALYSIS:\n")
KEY FINDINGS FROM PCA ANALYSIS:
> cat("1. Three principal components explain 55.3% of total variance\n")
1. Three principal components explain 55.3% of total variance
> cat("2. Component 1 (25.8%): Claims history vs. no-claims discount\n")
2. Component 1 (25.8%): Claims history vs. no-claims discount
> cat("3. Component 2 (17.2%): Vehicle power and usage patterns\n")
3. Component 2 (17.2%): Vehicle power and usage patterns
> cat("4. Component 3 (12.3%): Driver and vehicle age factors\n\n")
4. Component 3 (12.3%): Driver and vehicle age factors

> cat("RISK PROFILES IDENTIFIED:\n")
RISK PROFILES IDENTIFIED:
> cat("1. Young Power Drivers: ", nrow(young_power_drivers),
+     " policies\n", sep = "")
1. Young Power Drivers: 22 policies
> cat("2. High-Mileage Commercial: ", nrow(high_mileage_commercial),
+     " policies\n", sep = "")
2. High-Mileage Commercial: 54 policies
> cat("3. Aged Risk: ", nrow(aged_risk),
+     " policies\n", sep = "")
3. Aged Risk: 5 policies
> cat("4. Low Discount High Claim: ", nrow(low_discount_high_claim),
+     " policies\n\n", sep = "")
4.Low Discount High Claim: 71 policies

> cat("ACTUARIAL IMPLICATIONS:\n")
ACTUARIAL IMPLICATIONS:
```

```
> cat("1. Premium differentiation should consider these risk dimensions\n")
1. Premium differentiation should consider these risk dimensions
> cat("2. Underwriting guidelines should address specific risk profiles\n")
2. Underwriting guidelines should address specific risk profiles
> cat("3. Claims prevention programs could target high-risk groups\n")
3. Claims prevention programs could target high-risk groups
> cat("4. Portfolio monitoring should track evolution of risk profiles\n")
4. Portfolio monitoring should track evolution of risk profiles
> # END OF PROBLEM 1 ANALYSIS
>cat("\n==============================================================\
n")
==============================================================
> cat("END OF PROBLEM 1 ANALYSIS\n")
END OF PROBLEM 1 ANALYSIS
>cat("==============================================================\n"
)
```

APPENDIX B

```
#Problem_2:ASSOCIATION BETWEEN CATEGORICAL VARIABLES
>cat("\n==================================================================\
n")
==================================================================
> cat("START OF PROBLEM 2 ANALYSIS\n")
START OF PROBLEM 2 ANALYSIS
>cat("==================================================================\n"
)
==================================================================
> #creating contingency table with totals
> observed <- table(data1$claim_type, data1$transmission)
> addmargins(observed)

            Automatic Manual  Sum
  BodilyInjury        94    112  206
  Other               43     72  115
  Property           253    333  586
  Windscreen         150    143  293
  Sum                540    660 1200
> #let test be the chi square test
> test <- chisq.test(observed)
> #find the expected counts
> expected_counts<- test$expected
> print(expected_counts)

            Automatic Manual
  BodilyInjury     92.70 113.30
  Other            51.75  63.25
  Property        263.70 322.30
```

```
  Windscreen      131.85 161.15
>
> #let stat be the test statistic and df be the test parameter
> stat <- test$statistic
> df <- test$parameter
> critical_value <- qchisq(0.95, df)
> chisq.test(observed)

        Pearson's Chi-squared test

data:  observed
X-squared = 8.0551, df = 3, p-value = 0.04489
> ##creating the chi-square graph
> #PLOT 8: Chi-squared Distribution
> curve(dchisq(x, df), from = 0, to = qchisq(0.999, df),
+       main = 'Chi-Squared Distribution(df = 3)',
+       ylab = 'Density',
+       lwd = 2)
> #creating vector of x values
> critical_value <- qchisq(0.95, df)
> x_vector <- seq(from = critical_value, to = qchisq(0.999, df), length.out = 100)
> p_vector <- dchisq(x_vector, df)
> polygon(c(x_vector, rev(x_vector)), c(p_vector, rep(0, length(p_vector))), col =
adjustcolor('red', alpha=0.3), border = NA)
> abline(v = stat, col = "blue", lwd = 2, lty = 2)
> text(x = stat + 2.5 , y = 0.05, labels = paste("Test Statistic =", round(stat, 4)), col = "blue")
> abline(v = critical_value, col = "darkgreen", lwd = 2, lty = 2)
> text(x = critical_value - 2.5, y = 0.08, labels = paste("Critical value =",
round(critical_value, 4)), col = "darkgreen")
>
> #standardised residuals
```

```
> test$stdres

             Automatic     Manual
  BodilyInjury  0.2000413 -0.2000413
  Other        -1.7248339  1.7248339
  Property     -1.2420921  1.2420921
  Windscreen    2.4515581 -2.4515581
> #End of Task 2
>cat("\n==================================================================\
n")
==================================================================
> cat("END OF PROBLEM 2 ANALYSIS\n")
END OF PROBLEM 2 ANALYSIS
>cat("==================================================================\n"
)
==================================================================
> print(sessionInfo())
R version 4.5.3 (2026-03-11 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=English_Mauritius.utf8  LC_CTYPE=English_Mauritius.utf8
LC_MONETARY=English_Mauritius.utf8
[4] LC_NUMERIC=C                       LC_TIME=English_Mauritius.utf8

time zone: Indian/Mauritius
tzcode source: internal
```

```
attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] nFactors_2.4.1.2 ggpubr_0.6.3     naniar_1.1.0     psych_2.6.3      gridExtra_2.3
corrplot_0.95    factoextra_2.0.0
 [8] FactoMineR_2.13  lubridate_1.9.5  forcats_1.0.1    stringr_1.6.0    dplyr_1.2.0
purrr_1.2.1      readr_2.2.0
[15] tidyr_1.3.2      tibble_3.3.1     ggplot2_4.0.2    tidyverse_2.0.0

loaded via a namespace (and not attached):
 [1] gtable_0.3.6        htmlwidgets_1.6.4   visdat_0.6.0        ggrepel_0.9.8       rstatix_0.7.3
lattice_0.22-9
 [7] tzdb_0.5.0          vctrs_0.7.2         tools_4.5.3         generics_0.1.4      parallel_4.5.3
cluster_2.1.8.2
[13] pkgconfig_2.0.3     RColorBrewer_1.1-3  S7_0.2.1            scatterplot3d_0.3-45
lifecycle_1.0.5      compiler_4.5.3
[19] farver_2.1.2        mnormt_2.1.2        leaps_3.2           carData_3.0-6
htmltools_0.5.9      Formula_1.2-5
[25] car_3.1-5           pillar_1.11.1       MASS_7.3-65         flashClust_1.1-4     DT_0.34.0
abind_1.4-8
[31] nlme_3.1-168        tidyselect_1.2.1    digest_0.6.39       mvtnorm_1.3-6
stringi_1.8.7        labeling_0.4.3
[37] fastmap_1.2.0       grid_4.5.3          cli_3.6.5           magrittr_2.0.4       broom_1.0.12
withr_3.0.2
[43] scales_1.4.0        backports_1.5.0     estimability_1.5.1  timechange_0.4.0
emmeans_2.0.2        ggsignif_0.6.4
[49] hms_1.1.4           rlang_1.1.7         Rcpp_1.1.1          glue_1.8.0           rstudioapi_0.18.0
R6_2.6.1
```